

# Scrambling for Power: The Impact of Sudden Reactor Shutdowns on Short-Run Electricity Prices

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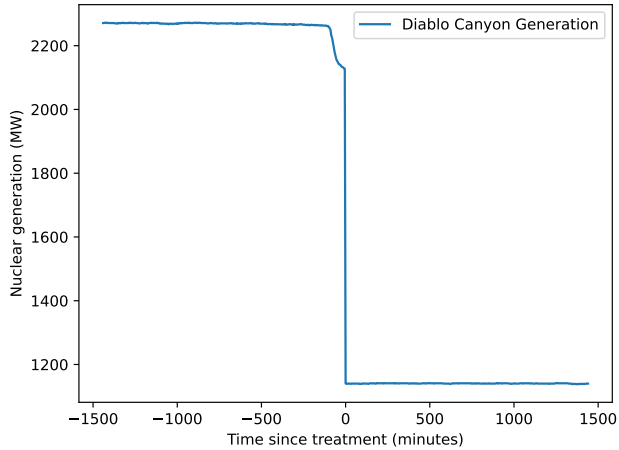
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- Nuclear reactors (famously) sometimes suffer mechanical/electrical faults and have to immediately shutdown
  - "Scrams"
- Vast majority of scrams have no broader impact besides the reactor going offline for a few hours/day
  - Key feature of scrams are that they happen almost instantaneously, sending production to 0
- Since reactors usually produce  $\sim 1$  GW, impact on power is significant
  - For context: 1 GW is roughly 2 million solar panels or 300 wind turbines
- When reactors go offline we can think of this as a massive supply shock with corresponding price effects

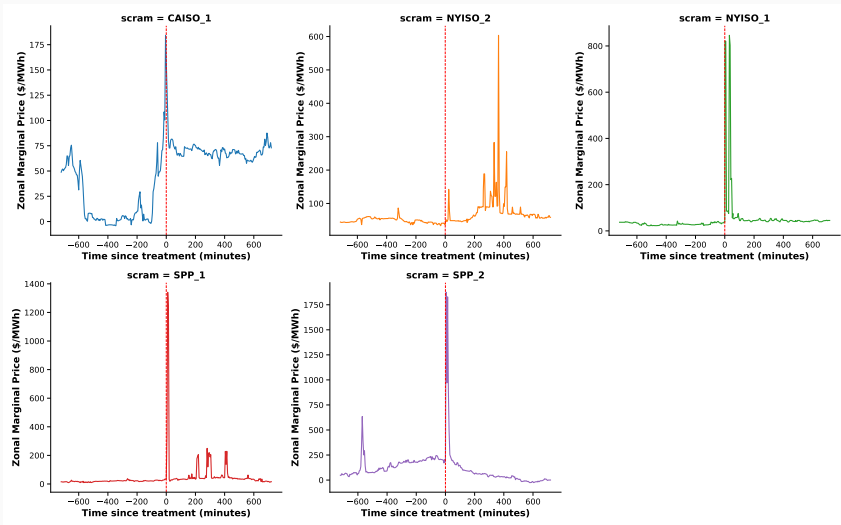
# Setting



# Why study this?

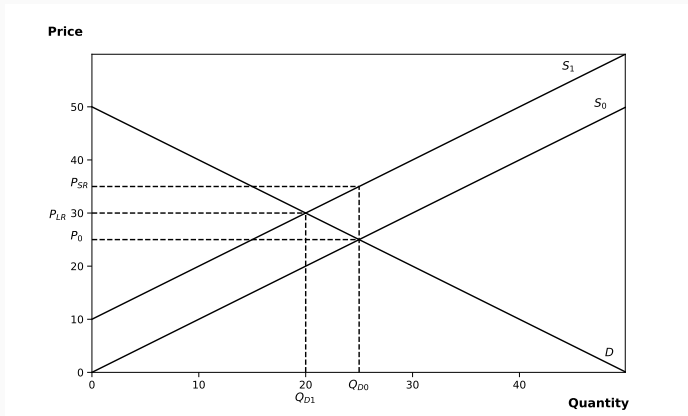
- Energy is incredibly important
  - Quintessential supply shock in macro literature
  - Interacts with every industry in some way
- Why focus on nuclear reactors?
  - Nuclear reactors are massive parts of the grid ( $\sim 20\%$  of total in the United States)
  - Beyond size, nuclear is "baseload" power
  - Baseload power is always on and balances out variable sources like solar and wind - especially important with green transition
- On the subject of green transition:
  - Nuclear is (almost) emissions free
  - Alternative sources of baseload power are typically natural gas, coal and hydro
  - Policymakers in the West love closing reactors yet there is still political will/actual need for decarbonization
- Goal of the study is to provide some evidence against closing more reactors by highlighting price shocks

# Why study this?



# Strategy

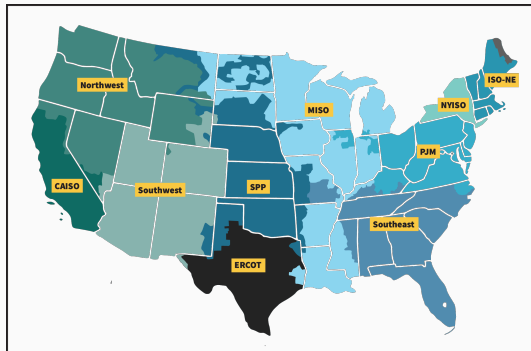
- Instantaneous supply shock naturally leads to a regression discontinuity in time (RDiT) approach
  - Key assumption is that we can think of demand and supply (outside of scram) as fixed in the short-run
  - Consumers can't shift consumption in advance since fault is by definition unexpected
  - Bulk of demand is commercial and industrial which have limited room to shift demand even after a scram - meaning  $Q_D$  adjusts only partially or not at all
  - Utilities can't shift production beyond imports or pushing more output from gas/coal - essentially the supply *curve* is fixed in the short run even as  $Q_S$  could be mostly made up
- Important result as simultaneity concerns complicate any analysis of a supply/demand shock
- In long-run equilibrium we expect to see a smaller effect as  $Q_D$  adjusts, but in short-run full effect size can be seen



**Figure 1:** Stylized supply and demand graph showing price and quantity scenarios.  $P_{SR}$  is short-run price,  $P_{LR}$  is the long-run price and  $P_0$  is the initial price.  $S_0$  and  $S_1$  are the pre- and post-scram supply curves.

# Sample and Data

- Sample: 13 scrams at 10 different plants
  - Each plant lives within an electricity market
  - I use CAISO, NYISO, SPP and PJM
  - Within each market, each plant lives within a zone
- Data was gathered from two sources
  - NRC Operating Scram database - I found prospective scrams in this database and checked them against...
  - EIA Wholesale Electricity Market data - contained price, generation and load data for the chosen markets



**Figure 2:** Geographical coverage of U.S. electricity markets

# Model

- We run the following regression

$$P_{it} = \alpha_j + \lambda_{hour} + \theta_b b_{it} + \theta_a a_{it} + \rho \mathbb{1}_{scram} \times D_i + \beta_1 Load_{it} + \epsilon_{it}$$

- $P_{it}$  = Zonal marginal price (ZMP) in \$/MWh
- $\alpha_j$  = Scram level fixed effect
- $\lambda_{hour}$  = hour of day fixed effect
- $b_{it}, a_{it}$  = Time before/after treatment
- $\rho$  = Effect of 1 MW lost on ZMP
- $\mathbb{1}_{scram} \times D_i$  = Indicator for whether the reactor scrammed yet interacted with the amount of MW lost from the scram
- $Load_{it}$  = Real time demand on the grid in MW
- Errors are clustered at the market level to account for correlated shocks
- $t$  is relative to treatment time, with the bandwidth being 12 hours before and after treatment

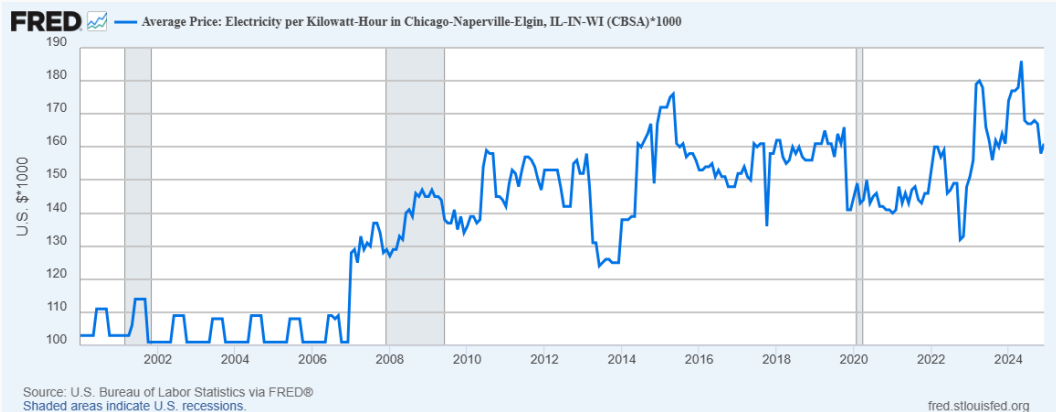
**Table 1:** Estimates of  $\rho$  by Specification and Fixed Effects

| Specification | OLS                  | Scram FE            | Plant FE            |
|---------------|----------------------|---------------------|---------------------|
| Load Only     | −0.012805 (0.013824) | 0.026176 (0.007683) | 0.027106 (0.007338) |
| No Controls   | −0.018373 (0.011784) | 0.027046 (0.010441) | 0.026447 (0.010460) |

*Notes:* Each cell reports  $\hat{\rho}$  with its standard error in parentheses. All specifications include time-fixed effects and are clustered at the market level. Observations: 1456 for OLS; 3,745 otherwise. OLS has less observations because PJM scrams were dropped since PJM generation data is too coarse to be useful.

- OLS of ZMP on nuclear generation shows a null result, as we would expect given simultaneity concerns discussed earlier
- Main RDiT specification shows  $\hat{\rho} = \$26.18 \pm 15.06/\text{MWh}$  per GW lost
- Quick robustness check: no controls and plant level fixed-effects don't change the effect size

# Effect Size



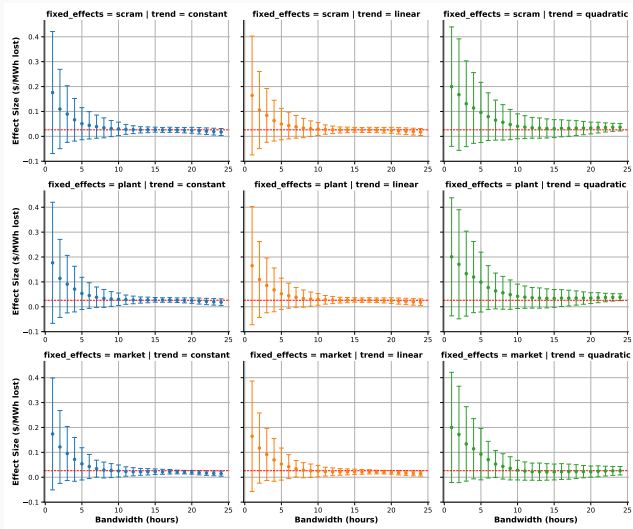
**Figure 3:** Electricity prices per MWh in the Chicago core-based statistical area. Price as of Dec 24 is roughly \$161/MWh

## Comparison to Other Studies

- Woo et. al (2014) find that the San Onofre shutdown in California raised prices by \$6-9/MWh with an estimated treatment effect of \$2.79-4.19/MWh per GW lost
- Rinne (2019) studied a widespread outage of nuclear reactors in France and found an effect size of 17.4 Euros per MWh or roughly \$19.74 per MWh giving an effect of \$8.60/MWh per GW lost
- Our effect size is moderately larger - why?
  - Woo explicitly argues supply/demand shifts soak up some of the effect
  - Both Woo and Rinne are over multiple month timeframes and compare pre- and post-treatment *equilibrium* prices
  - Our study only looks at immediate disequilibrium effect

- Possible alternate specifications:
  - Fixed effects at the market or plant level
  - No discontinuous running variable or locally quadratic term
  - Larger or smaller bandwidth
  - Controlling for temperature and/or dropping load
  - Placebo test (shifting treatment)
- Some of these tested in results slide

# Robustness

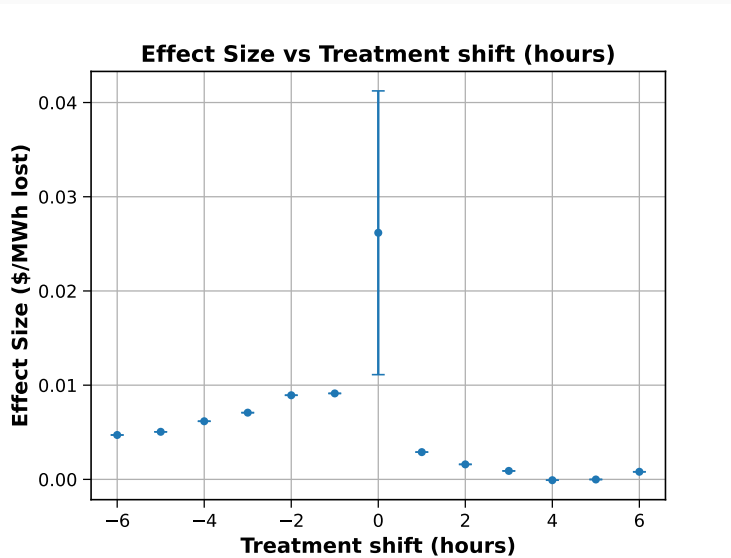


**Table 2:** Estimates of  $\rho$  by Specification and Fixed Effects

| Specification               | OLS                  | Scram FE            | Plant FE            |
|-----------------------------|----------------------|---------------------|---------------------|
| Standard Specification      | −0.012805 (0.013824) | 0.026176 (0.007683) | 0.027106 (0.007338) |
| Load & Temperature Controls | −0.015970 (0.013764) | 0.027608 (0.009907) | 0.026426 (0.007956) |
| Temperature Only            | −0.018373 (0.011784) | 0.031641 (0.013347) | 0.026162 (0.011915) |

*Notes:* Each cell reports  $\hat{\rho}$  with its standard error in parentheses. All specifications include time-fixed effects and are clustered at the market level. Observations: 2,914 for OLS; 3,745 otherwise. OLS has less observations because PJM scrams were dropped since PJM generation data is too coarse to be useful.

- Also ran a variance inflation factor (VIF) test and found the temperature covariates were  $\gg 10$  which indicated significant collinearity issues
- Time FE and load probably explain most of the variance in temperature anyway



# Conclusion

- RDiT approach exploits instantaneous drops in generation (scrams)
- Short-run window of interest gets around simultaneity issues and decreases downwards bias in effect size
- Effect is roughly \$26/MWh increase per GW lost of capacity
- Future directions
  - Include more scrams - data issues/time meant the sample was not super large
  - Heterogenous effects are definitely at play - quantifying what factors drive that heterogeneity could be extremely useful
  - More complicated import/exports model and other controls